

Unit 1. Probability Foundations

Sample spaces, events, axioms, counting, conditional probability and independence

Introductory Statistics for Econometrics

Universidad Carlos III de Madrid

Adapted from MIT OCW 14.30 (Spring 2009), K. Menzel, Lecture Notes 1–3, CC BY-NC-SA 4.0.

Roadmap

- 1 Random experiments and events
- 2 Set operations on events
- 3 Axioms of probability
- 4 Counting rules
- 5 Conditional probability
- 6 Independence
- 7 Law of Total Probability
- 8 Wrap-up

Why probability?

- Probability is a mathematical language to describe **situations with uncertainty**.
- In economics: asset returns, treatment effects, sampling, forecasts, identification.
- Today: the foundational vocabulary — *sample space, events, probability axioms, conditional probability, independence*.

Goal of Unit 1

Establish the basic objects and rules we will reuse in every later unit of this course — and in your econometrics courses.

Random experiment, sample space, events

Definitions

- A **random experiment** is a procedure that (a) can be repeated under identical conditions and (b) has a well-defined set of outcomes.
- The **sample space** $\mathcal{S}$ is the set of all possible outcomes.
- An **event** A is any subset $A \subseteq \mathcal{S}$.
- The **empty set** $\emptyset$ is defined as the event with no outcomes $\emptyset = \{\}$. Note $\emptyset \neq \{0\}$.

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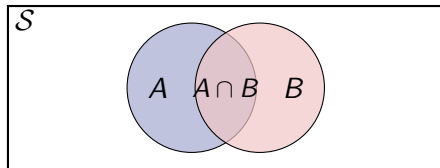
Example: rolling a die

- $\mathcal{S} = \{1, 2, 3, 4, 5, 6\}$.
- “The outcome is even” $= A = \{2, 4, 6\}$.
- “The outcome is at least 5” $= B = \{5, 6\}$.

Operations on events

For $A, B \subseteq \mathcal{S}$:

- **Union** $A \cup B$ — “*A or B*”.
- **Intersection** $A \cap B$ — “*A and B*”. We sometimes write AB .
- **Complement** $A^c = \mathcal{S} \setminus A$ — “*not A*”.
- **Disjoint** (mutually exclusive): $A \cap B = \emptyset$.
- $\{A_i\}$ is a **partition** of $\mathcal{S}$ if pairwise disjoint and $\bigcup_i A_i = \mathcal{S}$.



Useful identities

For all events $A, B, C \subseteq \mathcal{S}$

commutative: $A \cup B = B \cup A, \quad A \cap B = B \cap A,$

associative: $(A \cup B) \cup C = A \cup (B \cup C),$

distributive: $A \cap (B \cup C) = (A \cap B) \cup (A \cap C),$

De Morgan: $(A \cup B)^c = A^c \cap B^c,$

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- De Morgan's laws will be used constantly when computing probabilities of complements (“not all $\Leftrightarrow$ at least one is not”).

Kolmogorov's axioms

A **probability** is a function $\mathbb{P} : \mathcal{A} \rightarrow \mathbb{R}$ satisfying:

Axioms

(P1) $\mathbb{P}(A) \geq 0$ for every event A .

(P2) $\mathbb{P}(\mathcal{S}) = 1$.

(P3) If $A_1, A_2, \dots$ are pairwise disjoint, then $\mathbb{P}\left(\bigcup_i A_i\right) = \sum_i \mathbb{P}(A_i)$.

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These are **minimal** requirements: they don't single out a particular $\mathbb{P}$.

The art of statistics is then twofold:

- 1 choose / postulate a probability model;
- 2 check whether it describes the data.

Consequences of the axioms

Proposition

For any events A, B :

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Sketch of (1). $S = A \cup A^c$, $A \cap A^c = \emptyset$, so by (P2) and (P3): $1 = \mathbb{P}(S) = \mathbb{P}(A) + \mathbb{P}(A^c)$. $\square$

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When $\mathcal{S}$ is finite and outcomes are symmetric

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The probability question reduces to a **counting** problem.

Example. Roll one fair die.

$$\mathbb{P}(\text{outcome} > 4) = \frac{\#\{5, 6\}}{\#\{1, \dots, 6\}} = \frac{2}{6} = \frac{1}{3}.$$

Three counting rules

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- ④ **Combinations** (without replacement, unordered):

$$\binom{N}{n} = \frac{N!}{(N-n)! n!}.$$

Birthday paradox

n people, uniform birthdays:

$$\mathbb{P}(\text{at least two share a birthday}) = 1 - \frac{365!}{(365 - n)! 365^n}.$$

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Lesson for econometrics. The number of *potential* coincidences grows much faster than the number of observations — a warning about data mining and false discoveries.

Conditional probability

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Intuition. The known event B becomes the new sample space; the numerator restricts A to B , the denominator re-normalises.

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- Conditional probabilities will reappear as **conditional distributions**, the **conditional expectation function** and as **Bayesian updating** in later units.

Independence

Definition

Events A and B are **independent**, $A \perp B$, if

$$\mathbb{P}(A \cap B) = \mathbb{P}(A) \mathbb{P}(B).$$

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- Independence is a *property of the probability distribution*, not of the physical setup.
- If $A \perp B$, then $A \perp B^c$, $A^c \perp B$ and $A^c \perp B^c$.
- Pairwise independence does **not** imply mutual independence.

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A and B are independent *conditional on* C (with $\mathbb{P}(C) > 0$) if

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Crucial warning.

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This subtlety is the formal content of econometric phrases such as “*controlling for* X ” or “*ceteris paribus*”.

Law of Total Probability

Theorem

Let $\{B_1, \dots, B_n\}$ be a partition of $\mathcal{S}$ with $\mathbb{P}(B_i) > 0$. For any event A :

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Why we care.

- It decomposes a marginal probability into a weighted average of conditional probabilities.
- This is the discrete prototype of the **law of iterated expectations** $\mathbb{E}[Y] = \mathbb{E}[\mathbb{E}[Y \mid X]]$ that you will use in econometrics every week.

Composition effects (Simpson's paradox)

Within each procedure (partition B_i), experienced surgeons have *lower* mortality. But aggregated across procedures, their mortality is *higher*.

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	death rate	% of cases	death rate	% of cases
Single bypass	4.0%	50%	2.0%	25%
Double bypass	6.0%	40%	4.0%	25%
Triple bypass	10.0%	9%	6.0%	25%
Quadruple bypass	20.0%	1%	12.0%	25%
Marginal	5.5%		6.0%	

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Reading. The weights $\mathbb{P}(B_i)$ differ across groups. Disentangling this is the central task of regression analysis.

Take-aways

- 1 A probability model is a triple $(\mathcal{S}, \mathcal{A}, \mathbb{P})$ obeying Kolmogorov's axioms.
- 2 Set algebra (union, intersection, complement, De Morgan) is the algebra of events.
- 3 Classical probability = counting: multiplication, permutations, combinations.
- 4 Conditional probability rescales by the conditioning event; independence $\Leftrightarrow \mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$.
- 5 The Law of Total Probability is the building block for conditional expectations and regression.

Next unit

Random variables, probability mass / density functions, and cumulative distribution functions.

Thank you

Questions?

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